

Counting Colored Linear Snake Polyominoes Up to Symmetry

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Abstract

I count colored snake polyominoes in their linear arrangement: L cells in a straight line, each with binary edge colors and degree constraint $\sum \leq 2$. The single linear shape isolates the coloring problem. A 2-state transfer matrix yields raw counts; a Burnside-style average over the line stabilizer (with global flip) gives orbit counts. The worked example $L = 2$ produces 65 raw colorings and 21 distinct orbits. The linear orbit sequence starts 4, 21, 109, 586, 3326, $\dots$ and matches brute force and project tests through $L = 15$.

1 Linear model

Shape: L consecutive unit squares on a line. Each cell has edge bits $(S, E, N, W) \in \{0, 1\}^4$ with degree constraint $\sum \leq 2$, giving 11 admissible patterns. Adjacent cells must agree on the shared hinge edge. With geometry fixed, the task reduces to counting colorings and collapsing symmetries.

2 Symmetry

For $L > 1$ the geometric stabilizer G_0 has 4 elements (identity, path reversal, reflection across the line, reflection composed with reversal); for $L = 1$, $|G_0| = 8$. Adding global color flip yields acting group $G = G_0 \times \{\text{id}, \text{flip}\}$. Distinct linear snakes are orbits of colorings under G .

3 Transfer matrix

Only the hinge bit (0 or 1) passes between cells. Over the 11 admissible patterns, the West→East transfer matrix is

$$M = \begin{pmatrix} 4 & 3 \\ 3 & 1 \end{pmatrix},$$

rows = entry bit, columns = exit bit. Endpoint vectors are $v_{\text{in}} = v_{\text{out}} = (7, 4)$. Raw colorings:

$$\text{Raw}(L) = v_{\text{in}} M^{L-2} v_{\text{out}}^\top, \quad \text{Raw}(1) = 11.$$

4 Burnside over the line stabilizer

Let $\text{Fix}_V(g)$ be colorings fixed by g with all 11 patterns; $\text{Fix}_{\text{both}}(g)$ restricts to degree-2 patterns (needed for the flip terms). The orbit count is

$$N(L) = \frac{2 \sum_{g \in G_0} |\text{Fix}_V(g)| - \sum_{g \in G_0} |\text{Fix}_{\text{both}}(g)| + \sum_{g \in G_0} |\text{Fix}_{\text{both}}(f \circ g)|}{2 |G_0|},$$

with f the global flip. For $L > 1$, $|G_0| = 4$; for $L = 1$, $|G_0| = 8$.

5 Worked example: $L = 2$

- Raw: $\text{Raw}(2) = 65$.
- $|G_0| = 4$, full group size 8.
- Burnside numerator yields $N(2) = 21$.

So 21 distinct linear colored snakes of length 2.

6 Linear sequence

Orbit counts (transfer + Burnside) for the linear arrangement:

4, 21, 109, 586, 3326, 19209, 111871, 653758,
3824678, 22387074, 131052313, 767211817, 4491420695, 26293679325, 153927402355 ($L = 1..15$).

Raw counts (transfer only) start 11, 65, 380, 2196, 12804, $\dots$; orbit counts are smaller after symmetry collapse.

7 Validation and cost

Tests cover: (i) brute-force orbit vs Burnside for $L \leq 3$; (ii) raw vs transfer for $L \leq 3$; (iii) Burnside numerator divisibility for $L \leq 8$. All pass. Computation for linear counts is dominated by fast powers of the 2×2 matrix; large L is cheap.

8 Beyond the line (context only)

For bent snakes, shapes are enumerated once with pruning; the same coloring + Burnside routine applies per shape, then totals are summed. Those results are secondary here; focus remains the linear sequence above.

9 Outlook

Next steps: degree-exact (sum=2) variants, weighted edges, and extracting characteristic polynomials to study linear-snake asymptotics.